



SEMESTER 2025/2026 (A251)

STTPK3023 SOFTWARE TESTING (A)

INDIVIDUAL ASSIGNMENT: NON-FUNCTIONAL TESTING

SUBMITTED TO:

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PREPARED BY:

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1. System Under Test (SUT)

The system used for our project is School Management System (SMS), a web-based system, utilizing languages of PHP and MySQL. This system has four roles, which are Admin, Teacher, Student, and Owner, and each role has different UIs and functionalities.

GitHub project link:

<https://github.com/ProjectsAndPrograms/school-management-system>

For this assignment, the non-functional testing will be focusing on the admin panel, where:

System: SMS (Admin Panel)

Key Features and Modules:

- User Authentication: Admin login with email and password.
- Student Management: Modify (Add, View, Edit, Delete) student's details.
- Multi-step Form: Student Details, Address, Emergency Contact.
- Dashboard: Analytics of teachers, students, latest notices, reminders.
- Search Functionality (In student dashboard): Search for student's name.

2. Non-Functional Testing Type

Non-functional testing chosen: Reliability Testing.

Definition: Reliability testing is testing to see the extent to which any software system continuously performs the specified functions over a period of time without failure.

Purpose:

- To ensure that the system operates consistently without crashes or pops up unexpected messages.
- To verify that multi-step processes (e.g., adding/editing teacher's or student's information) can be completed repeatedly without errors.

Real Project Application:

- Ensures that critical information, such as teacher and student data management does not fail during daily use.
- Confirms that multi-user entries remain stable over repeated tasks.

Importance for System:

- The admin panel is considered one of the critical aspects for school daily operations. Any failures in editing, or accessing student's and teacher's records will disrupt administrative tasks.
- Reliability testing will prevent failures in real-life usage by simulating repeated actions like login, data editing, and submission.

3. Software Metrics

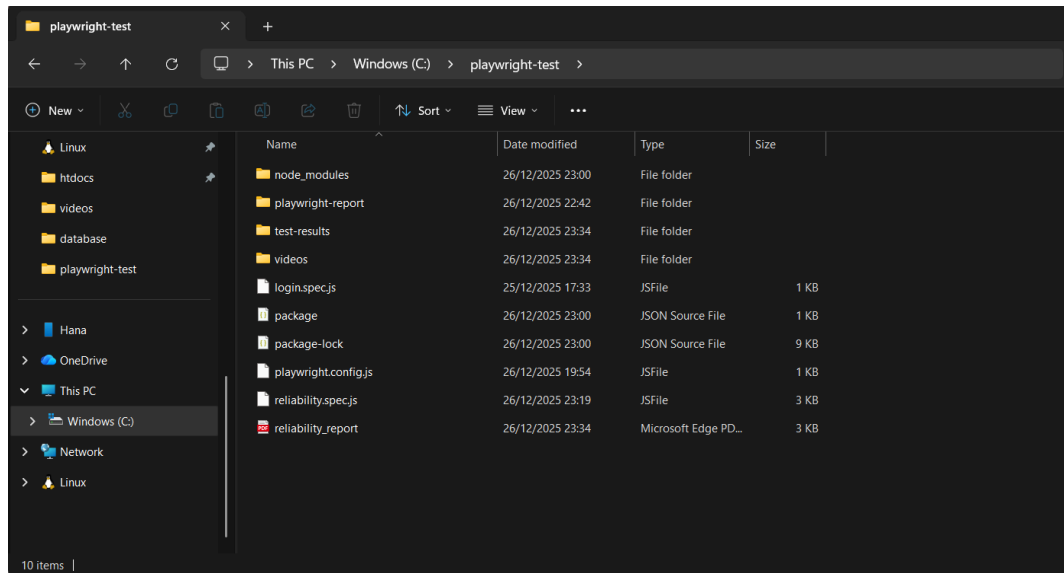
The software metrics for reliable testing are as follows:

Mean Time Between Failures (MTBF)	• Measure of the reliability of a system or component, representing the average time that a system or component will operate before it fails. • For this system, it's the average number of successful iterations that the admin can complete before the system fails (e.g. page crashes, timeout).
Recovery Time	• Recovery time is the time it takes for this system to return to a fully operational state after a failure. • For this system, it refers to the time it takes to restart the test browser and continue after a crash or timeout.
Failure Rate (%)	• Percentage of operations that fail out of the total operations executed. • For this system, it refers to the percentage of iterations where the admin edit flow did not complete successfully.

4. Automated Testing Tool

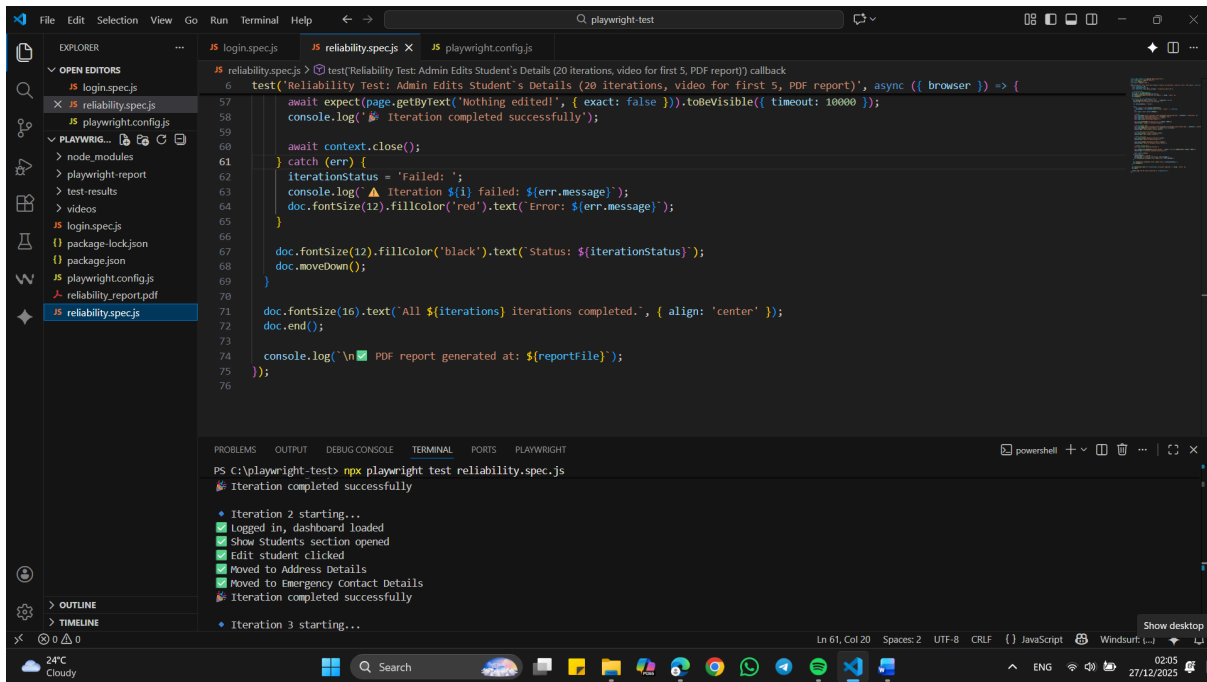
Automated Testing Tool Chosen: Playwright (Node.js)

Screenshots of setup:



- Created an empty folder under C:\ named playwright-test
- Opened the empty folder at VSCode
- Setup Playwright at terminal:

```
npm init -y
npm install -D @playwright/test
npx playwright install
```



- Setup at VSCode: Node.js environment with Playwright installed.
- Test script configured to perform login → navigate to Student page → edit student details → submit form → verify confirmation message.
- Configured for 20 iterations, with video recording for the first 5.

Test script: [reliability.spec.js](#)

```

const { test, expect } = require('@playwright/test');
const PDFDocument = require('pdfkit');
const fs = require('fs');
const path = require('path');

test('Reliability Test: Admin Edits Student's Details (20 iterations, video for first 5, PDF report)', async ({ browser }) => {
  const iterations = 20;
  const videoIterations = 5;
  const reportFile = path.join(__dirname, 'reliability_report.pdf');

  // Initialize PDF
  const doc = new PDFDocument();
  doc.pipe(fs.createWriteStream(reportFile));
  doc.fontSize(20).text('Reliability Test Report', { align: 'center' });
  doc.moveDown();

  for (let i = 1; i <= iterations; i++) {
    doc.fontSize(14).text(`Iteration ${i}:`, { underline: true });
    console.log(`\n Iteration ${i} starting...`);

    let iterationStatus = 'Passed';

    try {
      const context = await browser.newContext({
        recordVideo: i <= videoIterations ? { dir: 'videos/' } : undefined,
      });
      const page = await context.newPage();

```

```

// Login
await page.goto('http://localhost/school-management-system/login.php', {
waitUntil: 'networkidle' });
await page.fill('input[type="email"]', 'admin@gmail.com');
await page.fill('input[type="password"]', '123');
await page.click('button[type="submit"]');

await page.waitForURL(/dashboard\.php/, { timeout: 20000 });
console.log('✅ Logged in, dashboard loaded');

// Go to Student page
await
page.goto('http://localhost/school-management-system/admin_panel/student.php', {
waitUntil: 'networkidle' });
await page.getByText('Show Students', { exact: false }).click();
console.log('✅ Show Students section opened');

// Click Edit for first student
await page.locator('a.edit').first().click();
console.log('✅ Edit student clicked');

// Multi-step form
await page.locator('#general-info-btn').click();
console.log('✅ Moved to Address Details');

await page.locator('#personal-info-btn').click();
console.log('✅ Moved to Emergency Contact Details');

// Submit without edits
await page.click('#guardian-form-btn');

await expect(page.getByText('Nothing edited!', { exact: false })).toBeVisible({
timeout: 10000 });
console.log('Iteration completed successfully');

await context.close();
} catch (err) {
iterationStatus = 'Failed: ';
console.log(`⚠️ Iteration ${i} failed: ${err.message}`);
doc.fontSize(12).fillColor('red').text(`Error: ${err.message}`);
}

doc.fontSize(12).fillColor('black').text(`Status: ${iterationStatus}`);
doc.moveDown();
}

doc.fontSize(16).text(`All ${iterations} iterations completed.`, { align: 'center' });
doc.end();

console.log(`\n PDF report generated at: ${reportFile}`);
});

```

5. Test Scenario

Scenario	Test reliability of Admin editing student details function by repeatedly doing the same workflow over multiple iterations to observe system stability.
Scenario Description	Admin logs in to SMS, navigates to student dashboard, opens existing

	student record, proceeds through multi-step edit, and submit the form. This scenario is repeated 20 times sequentially to simulate repeated real-world usage.
User Flow	1. Admin logs into the system 2. System redirects to dashboard 3. Admin navigates to Student → Show Students 4. Admin selects a student and clicks Edit 5. Admin proceeds through: ○ Student Details ○ Address Details ○ Emergency Contact Details 6. Admin submits the form without changes 7. System displays confirmation message (“Nothing edited!”)
Endpoints	● /login.php ● /admin_panel/dashboard.php ● /admin_panel/student.php
Tool Used	Playwright, to automate browser-user based interactions and repeatedly execute the same test flow.
Test Configuration	● Number of iterations: 20 ● Browser contexts: New browser context per iteration ● Video recording: Enabled for first 5 iterations ● Execution type: Sequential (single user) ● Environment: Localhost (XAMPP)
Metrics	● Failure Rate (%) ● Mean Time Between Failures (MTBF) ● Recovery Time

6. Test Execution & Results

- Configuration:

```
npx playwright test reliability.spec.js
```

- First five iterations are recorded as videos.

Results (from terminal during execution):

```
Running 1 test using 1 worker
```

```
1 reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report)
```

```
  ♦ Iteration 1 starting...
  ✔ Logged in, dashboard loaded
  ✔ Show Students section opened
  ✔ Edit student clicked
  ✔ Moved to Address Details
  ✔ Moved to Emergency Contact Details
  Iteration completed successfully
```

```
  ♦ Iteration 2 starting...
  ✔ Logged in, dashboard loaded
  ✔ Show Students section opened
  ✔ Edit student clicked
  ✔ Moved to Address Details
  ✔ Moved to Emergency Contact Details
  Iteration completed successfully
```

```
  ♦ Iteration 3 starting...
  ✔ Logged in, dashboard loaded
  ✔ Show Students section opened
  ✔ Edit student clicked
  ✔ Moved to Address Details
  ✔ Moved to Emergency Contact Details
  Iteration completed successfully
```

```
  ♦ Iteration 4 starting...
  ✔ Logged in, dashboard loaded
  ✔ Show Students section opened
  ✔ Edit student clicked
  ✔ Moved to Address Details
  ✔ Moved to Emergency Contact Details
  Iteration completed successfully
```

```
  ♦ Iteration 5 starting...
  ✔ Logged in, dashboard loaded
  ✔ Show Students section opened
  ✔ Edit student clicked
  ✔ Moved to Address Details
  ✔ Moved to Emergency Contact Details
  Iteration completed successfully
```

```
  ♦ Iteration 6 starting...
  ✔ Logged in, dashboard loaded
  ✔ Show Students section opened
  ✔ Edit student clicked
```


✓ Moved to Address Details
✓ Moved to Emergency Contact Details
Iteration completed successfully

♦ Iteration 7 starting...
✓ Logged in, dashboard loaded
✓ Show Students section opened
✓ Edit student clicked
✓ Moved to Address Details
✓ Moved to Emergency Contact Details
Iteration completed successfully

♦ Iteration 8 starting...
✓ Logged in, dashboard loaded
✓ Show Students section opened
✓ Edit student clicked
✓ Moved to Address Details
✓ Moved to Emergency Contact Details
Iteration completed successfully

♦ Iteration 9 starting...
✓ Logged in, dashboard loaded
✓ Show Students section opened
✓ Edit student clicked
✓ Moved to Address Details
✓ Moved to Emergency Contact Details
Iteration completed successfully

♦ Iteration 10 starting...
✓ Logged in, dashboard loaded
✓ Show Students section opened
✓ Edit student clicked
✓ Moved to Address Details
✓ Moved to Emergency Contact Details
Iteration completed successfully

♦ Iteration 11 starting...
✓ Logged in, dashboard loaded
✓ Show Students section opened
✓ Edit student clicked
✓ Moved to Address Details
✓ Moved to Emergency Contact Details
Iteration completed successfully

♦ Iteration 12 starting...
✓ Logged in, dashboard loaded
✓ Show Students section opened
✓ Edit student clicked
✓ Moved to Address Details
✓ Moved to Emergency Contact Details
Iteration completed successfully

♦ Iteration 13 starting...
✓ Logged in, dashboard loaded
✓ Show Students section opened
✓ Edit student clicked
✓ Moved to Address Details
✓ Moved to Emergency Contact Details
Iteration completed successfully

♦ Iteration 14 starting...
✓ Logged in, dashboard loaded
⚠ Iteration 14 failed: page.goto: Test ended.
Call log:

- navigating to "http://localhost/school-management-system/admin_panel/student.php",

waiting until "networkidle"

♦ Iteration 15 starting...

⚠ Iteration 15 failed: browser.newContext: Test ended.

♦ Iteration 16 starting...

⚠ Iteration 16 failed: browser.newContext: Target page, context or browser has been closed

♦ Iteration 17 starting...

⚠ Iteration 17 failed: browser.newContext: Target page, context or browser has been closed

♦ Iteration 18 starting...

⚠ Iteration 18 failed: browser.newContext: Target page, context or browser has been closed

♦ Iteration 19 starting...

⚠ Iteration 19 failed: browser.newContext: Target page, context or browser has been closed

♦ Iteration 20 starting...

⚠ Iteration 20 failed: browser.newContext: Target page, context or browser has been closed

✅ PDF report generated at: C:\playwright-test\reliability_report.pdf

✖ 1 reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report) (1.0m)

1) reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report)

Test timeout of 60000ms exceeded.

Error Context:

test-results\reliability-Reliability-Te-33696-deo-for-first-5-PDF-report-\error-context.md

1 failed

reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report)

PS C:\playwright-test>

✖ 1 reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report) (1.0m)

1) reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report)

Test timeout of 60000ms exceeded.

Error Context:

test-results\reliability-Reliability-Te-33696-deo-for-first-5-PDF-report-\error-context.md

1 failed

reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report)

✖ 1 reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report) (1.0m)

1) reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20

iterations, video for first 5, PDF report)

Test timeout of 60000ms exceeded.

✖ 1 reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report) (1.0m)

1) reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report)

✖ 1 reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report) (1.0m)

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✖ 1 reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report) (1.0m)

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1) reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report)

Test timeout of 60000ms exceeded.

Error Context:

test-results\reliability-Reliability-Te-33696-deo-for-first-5-PDF-report-\error-context.md

1 failed

reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report)

PS C:\playwright-test>

1) reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report)

Test timeout of 60000ms exceeded.

Error Context:

test-results\reliability-Reliability-Te-33696-deo-for-first-5-PDF-report-\error-context.md

1 failed

reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report)

1) reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20 iterations, video for first 5, PDF report)

Test timeout of 60000ms exceeded.

Error Context:

test-results\reliability-Reliability-Te-33696-deo-for-first-5-PDF-report-\error-context.
md

1 failed

reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20
iterations, video for first 5, PDF report)
PS C:\playwright-test>

1) reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20
iterations, video for first 5, PDF report)

Test timeout of 60000ms exceeded.

Error Context:

test-results\reliability-Reliability-Te-33696-deo-for-first-5-PDF-report-\error-context.
md

1 failed

reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20
iterations, video for first 5, PDF report)
PS C:\playwright-test>

1) reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20
iterations, video for first 5, PDF report)

Test timeout of 60000ms exceeded.

Error Context:

test-results\reliability-Reliability-Te-33696-deo-for-first-5-PDF-report-\error-context.
md

1 failed

reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20
iterations, video for first 5, PDF report)
PS C:\playwright-test>

Test timeout of 60000ms exceeded.

Error Context:

test-results\reliability-Reliability-Te-33696-deo-for-first-5-PDF-report-\error-context.
md

1 failed

reliability.spec.js:6:1 › Reliability Test: Admin Edits Student`s Details (20
iterations, video for first 5, PDF report)

Report Generated by Tool:

Q	All 2	✓ Passed 1	✗ Failed 1	Flaky 0	Skipped 0		
--------------------------------	-------	------------	------------	---------	-----------	--	--

26/12/2025, 22:41:52 Total time: 1.1m

▼ reliability.spec.js	
✗ reliability test – admin edits student (20 iterations, video for first 5) reliability.spec.js:3	1.0m
▼ login.spec.js	
✓ open school management system login.spec.js:3	1.1s

Reliability Test Report

Iteration 1:

Status: Passed

Iteration 2:

Status: Passed

Iteration 3:

Status: Passed

Iteration 4:

Status: Passed

Iteration 5:

Status: Passed

Iteration 6:

Status: Passed

Iteration 7:

Status: Passed

Iteration 8:

Status: Passed

Iteration 9:

Status: Passed

Iteration 10:

Status: Passed

Iteration 11:

Status: Passed

Iteration 12:

Status: Passed

Iteration 13:

Status: Passed

Iteration 14:

Error: page.goto: Test ended.

Call log:
&[2m - navigating to "http://localhost/school-management-system/admin_panel/student.php", waiting until "networkidle"&[22m
Status: Failed:

Iteration 15:
Error: browser.newContext: Test ended.
Status: Failed:

Iteration 16:
Error: browser.newContext: Target page, context or browser has been closed
Status: Failed:

Iteration 17:
Error: browser.newContext: Target page, context or browser has been closed
Status: Failed:

Iteration 18:
Error: browser.newContext: Target page, context or browser has been closed
Status: Failed:

Iteration 19:
Error: browser.newContext: Target page, context or browser has been closed
Status: Failed:

Iteration 20:
Error: browser.newContext: Target page, context or browser has been closed
Status: Failed:

All 20 iterations completed.

Summary of results:

Iterations	Results
0-13	Pass
14-20	Failed due to timeout of system.

Metrics:

Mean Time Between Failures (MTBF)	Number of failures/Number of successful iterations = 13/7 ≈1.86 iterations per failure
Recovery Time	~6–7 seconds per failure
Failure Rate (%)	Total iterations/Failed iterations × 100 = 7/20 × 100 = 35%

7. Analysis & Discussion

1. Do the results meet expectations?
 - The results partially meet expectations. Iterations of 1 until 13 succeeded, showing the system is reliable for normal repeated use, but iterations 14-20 failed due to browser timeouts, showing stability when performing many consecutive iterations.
2. What part of the system is a bottleneck?
 - The login page and student navigation page after ~13 iterations, most likely due to CPU usage accumulation or session handling in XAMPP.
3. What improvements are recommended?
 - Optimize backend PHP scripts or database connections to handle repeated requests without timeouts.
 - Clear sessions or reset context between iterations to avoid resource accumulation.

8. Conclusion

- The system shows high reliability under repeated normal conditions (admin editing student details)
- Failures in iterations 14-20 shows a potential resource or session management issues.
- Automated testing using Playwright is effective for simulating realistic repeated user actions and generating iteration results.